

Individual Project 1 Report

WAPH-Web Application Programming and Hacking

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Student

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Short-bio: I am a Computer Science Junior at the University of Cincinnati with a strong foundation in AI/ML, data analytics, and edge computing. I specialize in building end-to-end projects.



Figure 1: Headshot

Individual Project 1 - Professional Profile Website

Overview

The objective of this project was to develop and deploy a professional front-end profile website on the `github.io` cloud service. This assignment expanded my front-end development skills by requiring the use of CSS frameworks for a professional appearance, integration of public Web APIs to create dynamic content, and the implementation of JavaScript cookies to track user sessions.

Through this project, I learned how to manage front-end dependencies, handle asynchronous API requests, and persist data in a client-side environment.

- **Live Website URL:** <https://nagpalvv.github.io/>
- **GitHub Repository:** <https://github.com/nagpalvv/waph-nagpalvv.github.io>

Tasks and Sub-tasks

1. Professional Profile and Deployment

- **Summary:** Created a responsive professional website using the “Start Bootstrap” resume theme. Included my background, education, and professional experience, as well as a linked `waph.html` course page.



2. Technical Requirements: Clocks and Web API Integration

- **Summary:** Integrated a digital and analog clock for real-time display and configured two public Web APIs:
 - **JokeAPI:** Fetches and displays a new programming joke every 60 seconds.
 - **Dog API:** Dynamically fetches and displays a random dog graphic.
 - *Disclaimer:* Content generated by third-party APIs (JokeAPI/Dog.ceo). I am not responsible for the content.

WAPH Course Information

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Labs & Hackathons

- Lab 0: Development Environment Setup.
- Lab 1: Foundations of the Web (HTTP, CGI, PHP).
- Lab 2: Front-end Web Development (Ajax, jQuery, APIs).
- Hackathon 1: XSS Attacks and Defenses.

Technical Widgets

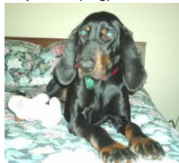
Time: 5:39:21 AM

API Integrations

Joke API:

A horse walks into a bar. "Hey", the Bartender says. "Sure", the horse replies.

Graphic API (Dog):



3. JavaScript Cookies

- **Summary:** Implemented `document.cookie` logic to detect first-time visitors and returning users, displaying a personalized “Welcome back!” message with the last recorded visit timestamp.

Conclusion

This project successfully demonstrated the integration of dynamic server-side data into a static front-end framework. By utilizing jQuery and modern `fetch()`

patterns, I improved my ability to manage asynchronous data streams while adhering to security best practices, such as sanitizing third-party content and managing client-side cookies.